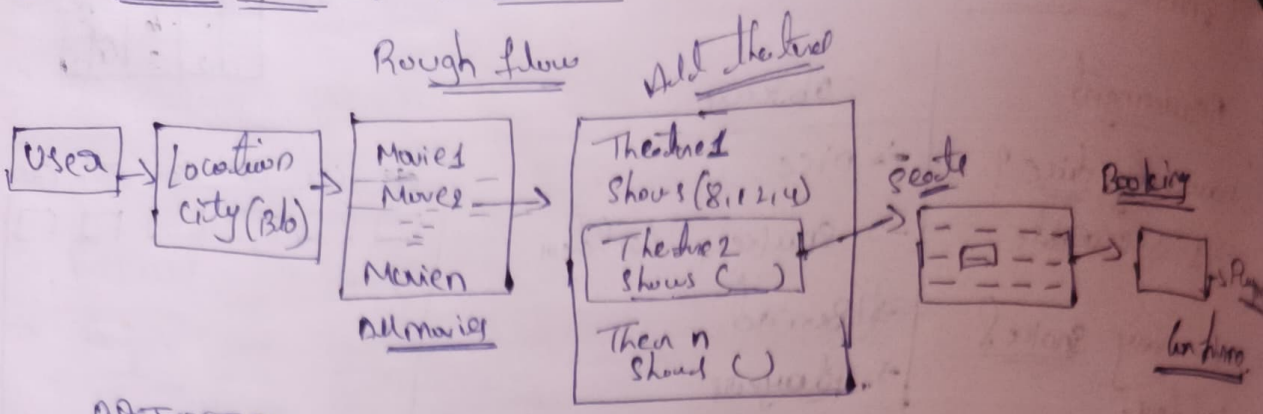
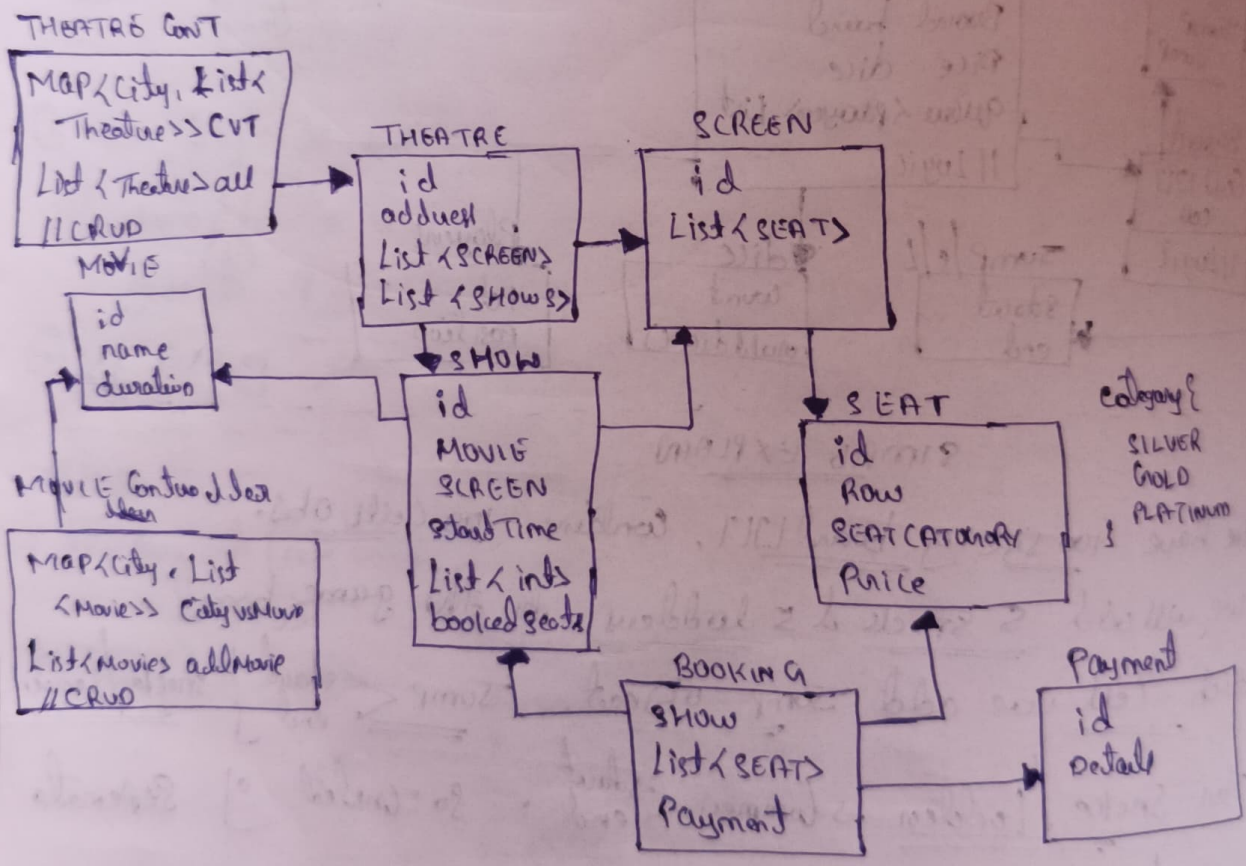
(7) Book My Show:- Concurrency.



OBJECTS

1. USER
2. MOVIE
3. CITY
4. THEATRE
5. SCREEN (halls)
6. SHOWS
7. SEAT
8. BOOKING
9. PAYMENT



1. Create MovieController & TheatreController to add Movies/Theatres.
2. Create Movies → add it using MovieController.
3. Create Theatre → SCREEN
   → SHOWS →
4. Create SCREEN → Create shows.

④ ADD all SHOWS.
⑤ SET List<Shows> to Theatre.
⑥ Theatre need → List SCREEN
List SHOWS

⑦ ADD Theatre using Theatre Controller

⑧ SCREEN → Need SEAT. → Create List SEAT. ADD Multiple SEAT OBJ
SET List SEAT.

CREATE BOOKING

① Get Movies by Location. (Call Movies by City).
② Based on Movie Name, get all [Theatre, List<SHOWS>] using Theatre Controller.
③ Choose Anyone SHOW on any Theatre in List<SHOWS>.
④ Based on SHOW, get all Booking SEATS & check given seat is available or NOT
⑤ Create Booking OBJ & set seat & SHOW.
⑥ If New user select same, he cannot Book, since booked seat has already Contain seat

→ we can also use Booked SEAT SET & Available SEAT.

CONCURRENCY → OPTIMISTIC
→ PESSIMISTIC

① OPTIMISTIC

→ Allow multiple USER to READ DATA.
→ During update, it checks (v) VERSION
If version matches both Read & update → Then update. else NO.

② PESSIMISTIC

→ Lock Row while Reading only. Does not allow all transaction to Read.
→ So, cannot READ & MODIFY by others.

CHOOSE

① USE OPTIMISTIC. better than PESSIMIST
② Not required to Lock full SEATS.

→ TIMER: Can be solved using "BUDDIE", set expiry time for Lock. automatic.